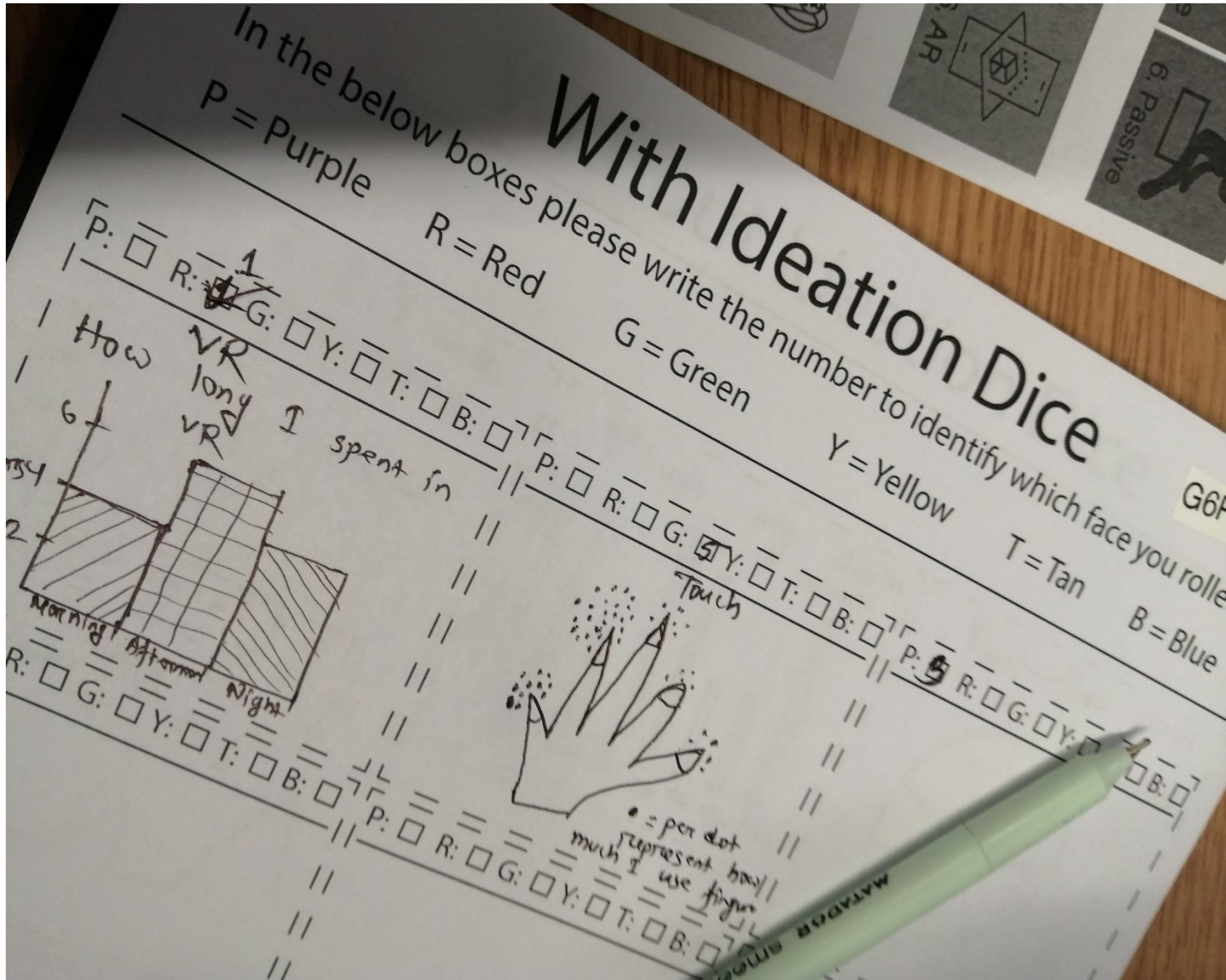


Inspire and Create: Unveiling the Potential of VisDice in Visualization Design

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We introduce dice to help with visualisation (VisDice): An innovative tool developed for the design process within visualization, serves as a catalyst for creative thought. By incorporating randomness, it prompts designers to explore novel ideas and alternative solutions, fostering inspiration and expanding creative horizons. This paper delves into the role of VisDice in stimulating creativity and investigates its potential impact on enhancing visualization outcomes.

Learn to ignite innovative thinking and broaden your creative boundaries with VisDice

students often possess limited awareness of various visualization types and design possibilities. frequently adhering to familiar approaches without delving into alternative options. This tendency can hinder their creative growth and result in missed opportunities to discover more effective and impactful visualization strategies.

Experts in the field exhibit a propensity to explore diverse visualization types and design variations due to their comprehensive understanding of the subject.

Even with this, novice designers often exhibit creative limitations from an inadequate grasp of the subject domain. Conversely, although veterans in the field may display a more comprehensive array of strategies, they could remain ensnared within particular cognitive frameworks, potentially stifling innovative ideation. This phenomenon resonates with the observations of Linus Pauling, a Nobel laureate in chemistry, who contended that prolific inspiration emerges from the generation of many ideas; as he famously remarked, "The best way to have a good idea is to have lots of ideas."

Iterative stages of VisDice

Stage 1 - Ideation

Designers initiate the process by rolling individual dice, each representing different design aspects, fostering a shift from traditional patterns of thinking.

The VisDice-driven ideation process starts with designers rolling distinct dice, each symbolizing specific design facets. This departure from tradition prompts exploration of new directions due to the introduced randomness. Dice cover diverse design elements like visual style and interaction methods, urging designers beyond their comfort zones. With each roll, varied combinations encourage adaptability and novel exploration. This shift cultivates imaginative concepts, acting as a catalyst for creativity and guiding designers towards innovative paths in visualization design.

Stage 2 – The Prompt

The rolled symbols on the dice provide unexpected and diverse prompts, challenging designers to contemplate new directions and possibilities.

As each symbol emerges, designers are prompted to engage in a cognitive shift, breaking away from routine thought patterns. This process of encountering unexpected visual cues serves as a catalyst for unearthing innovative design concepts that might have remained hidden within the confines of conventional ideation.

Stage 3 – Inspiration

Designers engage in ideation, leveraging the prompts to inspire creative thinking and formulate innovative design ideas.

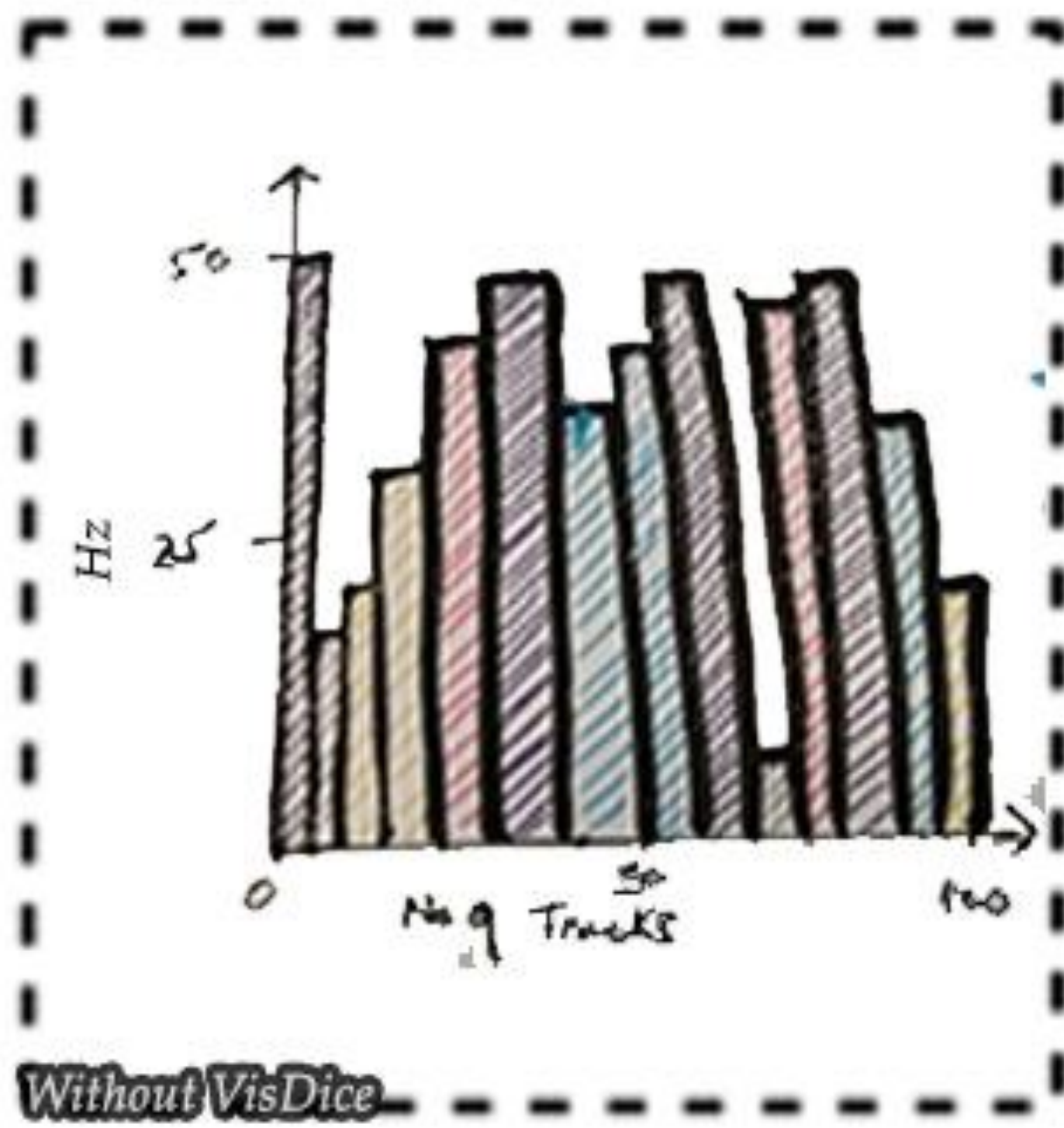
The interaction with VisDice nurtures a mindset of open-mindedness, enabling designers to break free from the constraints of routine thought and embrace the unfamiliar. Through this interaction, VisDice acts as a catalyst for inspiration, encouraging designers to harness their creative potential and craft solutions that are both novel and impactful.

Stage 4 – Creation

Based on the inspired ideation, designers proceed to create visualization sketches that embody the conceptualizations stemming from VisDice's prompts.

These sketches serve as tangible expressions of the innovative concepts that VisDice has ignited within the designers' minds. As designers translate these imaginative ideas into visual representations, they embark on a journey of transforming abstract thoughts into concrete design elements. The sketches become a canvas where the designers' creative visions take shape, merging the unpredictability of VisDice's prompts with their individual design sensibilities. This stage marks the transition from conceptualization to manifestation, capturing the essence of VisDice's influence on the creative process.

A. Learner



B.



Section A juxtaposes visualizations crafted without VisDice (on the left) and with (on the right) VisDice's impact, utilizing pitch data from the top 100 songs on Spotify. In contrast, Section B showcases the set of six meticulously crafted dice strategically engineered which aims to kindle innovative concepts and broaden the creative horizons of visualization designers.

Stage 5 – Evaluation

Designers critically assess the generated sketches, refining and iterating upon them to align with their creative vision and design objectives.

This phase involves a meticulous examination of the initial visualizations, where designers evaluate the harmonious integration of VisDice-inspired ideas with their overarching design goals. Through this critical evaluation, designers identify areas for enhancement, modification, and alignment with the intended message of the visualization.

Stage 6 – Finalisation

The iterative process culminates in the finalization of a visualization design that incorporates the innovative ideas and design concepts generated through the VisDice-driven ideation process.

At this juncture, designers amalgamate their refined sketches, weaving together a cohesive narrative that harmoniously integrates the unexpected insights sparked by VisDice. This synthesis marks the realization of the visualization's potential, as designers distill the multitude of ideational threads into a coherent and impactful whole.

